DATA 427 Group Project: Spark ML on the KDD Dataset

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1 Implement two programs that apply Spark's Decision Tree algorithm and Logistic Regression algorithm to the provided KDD dataset. Run these programs 10 times on the school's Hadoop cluster, each using a different seed to split the dataset into a training and a test set.

1.1 Describe the two programs using pseudo-code

1.1.1 Logistic Regression:

Algorithm 1 Spark Logistic Regression

```
Parse the input/output directories and number of runs
Initialise the Spark session and load the data from the provided input directory

Setup the pipeline as follows:
Index required columns
One-hot encode required columns
Assemble into feature columns
Append parameterised logistic regression model as the final stage.
for Number of runs do
    Set seed
    The data into train/test
    Fit the LR model on the train data
    Evaluate the models performance on the training and testing split, and record results
end for
Display results
```

1.1.2 Decision Tree

Algorithm 2 Spark Decision Tree

Parse the input/output directories and number of runs
Initialise the Spark session and load the data from the provided input directory
Generate schema
Setup the pipeline as follows:
Index required columns
One-hot encode required columns
Assemble into feature columns
Append parameterised logistic regression model as the final stage.
for Number of runs **do**
    Set seed
    The data into train/test
    Fit the Decision Tree odel on the train data
    Evaluate the models performance on the training and testing split, and record results
end for
Display results

1.2 Describe how to install and run the two programs step by step in a Readme.txt file.

*** README**

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*** Setup Notes**

- Ensure that you have a user directory on the HDFS cluster
- Download the Tarball into a folder on your ECS desktop
- Extract the tarball
- cd into the extracted folder

*** How to install**

If you are reading this, it is installed.

*** Directory Layout**

- `SparkWordCount/input` - the dataset extracted and ready for use
- `SparkWordCount/bin` - where the code is placed
- `SparkWordCount/staged_output` - where local runs place their output instead of HDFS
- `DecisionTree/input` - the dataset extracted and ready for use
- `DecisionTree/bin` - where the code is placed
- `DecisionTree/staged_output` - where local runs place their output instead of HDFS

- `LogisticRegression/input` - the dataset extracted and ready for use
- `LogisticRegression/bin` - where the code is placed
- `LogisticRegression/staged_output` - where local runs place their output instead of HDFS
- `ecs_hadoop_env` - the ECS cluster Hadoop makefile variables
- `ecs_spart_env` - the ECS cluster Spark makefile variables
- `local_hadoop_env` - the Hadoop makefile variables for local execution
- `local_spark_env` - the Hadoop makefile variables for local execution

*** How to run**

The makefile supports two execution modes, ecs and local, picked via a ENV environment variable. "ecs" is the default.

The makefile will ensure proper configured access to kerberos, HDFS and Spark before attempting to launch the job.

This is configured to run via makefile. To run a variant:

```
$ make submit_DecisionTree
$ make submit_SparkWordCount
$ make submit_LogisticRegression
```

or optionally

```
$ make all
```

The output will be placed in:

```
DecisionTree/output
LogisticRegression/output
SparkWordCount/output
```

```
$ cat DecisionTree/output/*.csv
$ cat LogisticRegression/output/*.csv
$ cat SparkWordCount/output/*.csv
```

To clean the files when complete:

```
$ make clean_DecisionTree
$ make clean_LogisticRegression
$ make clean_SparkWordCount
```

or

```
$ make clean
```

1.3 Report the training and test results including the max, min, average accuracy and the standard deviation obtained from the 10 runs, and the running time of each program.

Table 1: Summary of model test accuracy and total running time over 10 runs.

Program	Average acc. (%)	Std. dev.	Min acc. (%)	Max acc. (%)	Total time (s)
Decision Tree	94.57	0.20	94.37	95.06	65.37
Spark Logistic Regression	94.15	0.17	93.93	94.37	129.29